Delta Live Tables benefits and features

DLT is a sophisticated framework designed to construct reliable data pipelines. DLT automates and streamlines complex operations like orchestration and cluster management, significantly boosting the efficiency of data workflows. All you need to do is specify your transformation logic to be up and running. We will focus on a subset of the DLT's benefits as they pertain to creating the Bronze data layer, but we've included a link to DLT documentation in the *Further reading* section at the end of this chapter as well. DLT is a great tool in your toolbox for ingesting data because it provides several benefits over traditional ETL pipelines, including:

- **Declarative pipeline development**: Delta Live Tables allows you to define your data pipelines using SQL or Python, which makes them easier to understand and maintain.
- **Automatic data quality testing**: Delta Live Tables can automatically apply your tests to your data, preventing quality issues, which helps to ensure that your pipelines are producing accurate results.
- **Deep visibility for monitoring and recovery**: Delta Live Tables provides detailed tracking and logging information to make troubleshooting problems and recovering from failures easier.
- **Cost-effective streaming through efficient compute autoscaling**: Delta Live Tables can automatically scale your compute resources up or down based on demand, which helps to reduce costs.

Delta Live Tables is a powerful tool for building reliable and scalable data pipelines in batch or streaming. It can help you improve the quality, efficiency, and visibility of your data processing workflows.

Here are some of the specific features of Delta Live Tables that provide these benefits:

- **Streaming tables**: Delta Live Tables uses streaming and live tables to process data in real-time and to keep your pipelines up-to-date with the latest data.
- **Materialized views**: Delta Live Tables uses materialized views to create snapshots of your data. You can query your data in real-time and use it for downstream processing.
- **Expectations**: Delta Live Tables uses expectations to test your data for quality issues automatically and to take action if the data does not meet your expectations.
- **Autoscaling**: Delta Live Tables can automatically scale your compute resources up or down based on demand, reducing costs, and improving the performance of your pipelines.

Delta Live Tables is a good way to build reliable, scalable, and testable data pipelines. Next, we focus on DLT in the context of the Bronze layer.